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Java SE 27 & JDK 27

Class ReferenceQueue<T>

java.lang.Object
    java.lang.ref.ReferenceQueue<T>

Type Parameters:

T - the type of the reference object

```
public class ReferenceQueue<T>
extends Object
```

Reference queues, to which registered reference objects are appended by the garbage collector after the appropriate reachability changes are detected.

Memory consistency effects: The enqueueing of a reference to a queue (by the garbage collector, or by a successful call to `Reference.enqueue()`) *happens-before* the reference is removed from the queue by `poll()` or `remove(long)`.

Since:

1.2

Constructor Summary

Constructors

Constructor	Description
`ReferenceQueue()`	Constructs a new reference-object queue.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
`Reference<? extends T>`	`poll()`	Polls this queue to see if a reference object is available.
`Reference<? extends T>`	`remove()`	Removes the next reference object in this queue, blocking until one becomes available.
`Reference<? extends T>`	`remove(long timeout)`	Removes the next reference object in this queue, blocking until

	either one becomes available or the given timeout period expires.

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Constructor Details

ReferenceQueue

```
public ReferenceQueue()
```

Constructs a new reference-object queue.

Method Details

poll

```
public Reference<? extends T> poll()
```

Polls this queue to see if a reference object is available. If one is available without further delay then it is removed from the queue and returned. Otherwise this method immediately returns `null`.

Returns:

A reference object, if one was immediately available, otherwise `null`

See Also:

`Reference.enqueue()`

remove

```
public Reference<? extends T> remove(long timeout)
                              throws InterruptedException
```

Removes the next reference object in this queue, blocking until either one becomes available or the given timeout period expires.

This method does not offer real-time guarantees: It schedules the timeout as if by invoking the `Object.wait(long)` method.

Parameters:

`timeout` - If positive, block for up to `timeout` milliseconds while waiting for a reference to be added to this queue. If zero, block indefinitely.

Returns:

A reference object, if one was available within the specified timeout period, otherwise `null`

Throws:

`IllegalArgumentException` - If the value of the timeout argument is negative

`InterruptedException` - If the timeout wait is interrupted

See Also:

`Reference.enqueue()`

remove

```
public Reference<? extends T> remove()
                              throws InterruptedException
```

Removes the next reference object in this queue, blocking until one becomes available.

Returns:

A reference object, blocking until one becomes available

Throws:

`InterruptedException` - If the wait is interrupted

See Also:

`Reference.enqueue()`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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